

"Marlin EV: Front brake judder from warped front rotors"

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Area: bulletins

1 Condition

Owners of the Marlin EV may report a pulsation or judder felt through the brake pedal and steering wheel during moderate-to-heavy braking from highway speeds. The condition is most pronounced after sustained downhill braking or repeated stops and is caused by lateral runout of the front rotor exceeding specification. Inspect the front rotor assembly BR-7731 for thickness variation and scoring before proceeding. Refer to the Marlin EV Brakes Service Manual :: Front Pad and Rotor Specifications for the in-service runout and minimum thickness limits.

2 Cause

The judder results from uneven heat distribution across the front rotor BR-7731, which produces localized hard spots and thickness variation over time. Aggressive pad bed-in or extended regenerative-blending transitions can accelerate the formation of these hot spots on affected vehicles. The disc thickness variation eventually exceeds the published limit, transmitting a cyclic torque variation into the caliper and knuckle. Background on the front brake architecture is available in the Marlin EV Brakes Service Manual :: Front Brakes overview.

3 Repair Procedure

Perform the front rotor and pad service per [PROC-BRAKE-FR] as documented in the Marlin EV Brakes Service Manual :: Front Brake Pad and Rotor Service Procedure . Replace the front rotor BR-7731 in axle pairs and install a new front pad set BP-7720 to ensure even break-in and consistent friction characteristics. Torque the caliper bracket fasteners and confirm final lateral runout is within the value listed in the Marlin EV Brakes Service Manual :: Front Pad and Rotor Specifications . After reassembly, complete the bed-in cycle and verify the judder is no longer present; consult the full Marlin EV Brakes Service Manual for torque tables and bleed sequencing.